VELS VAC – Python to Generative AI Bootcamp

Day 1 – Foundations of Python & AI

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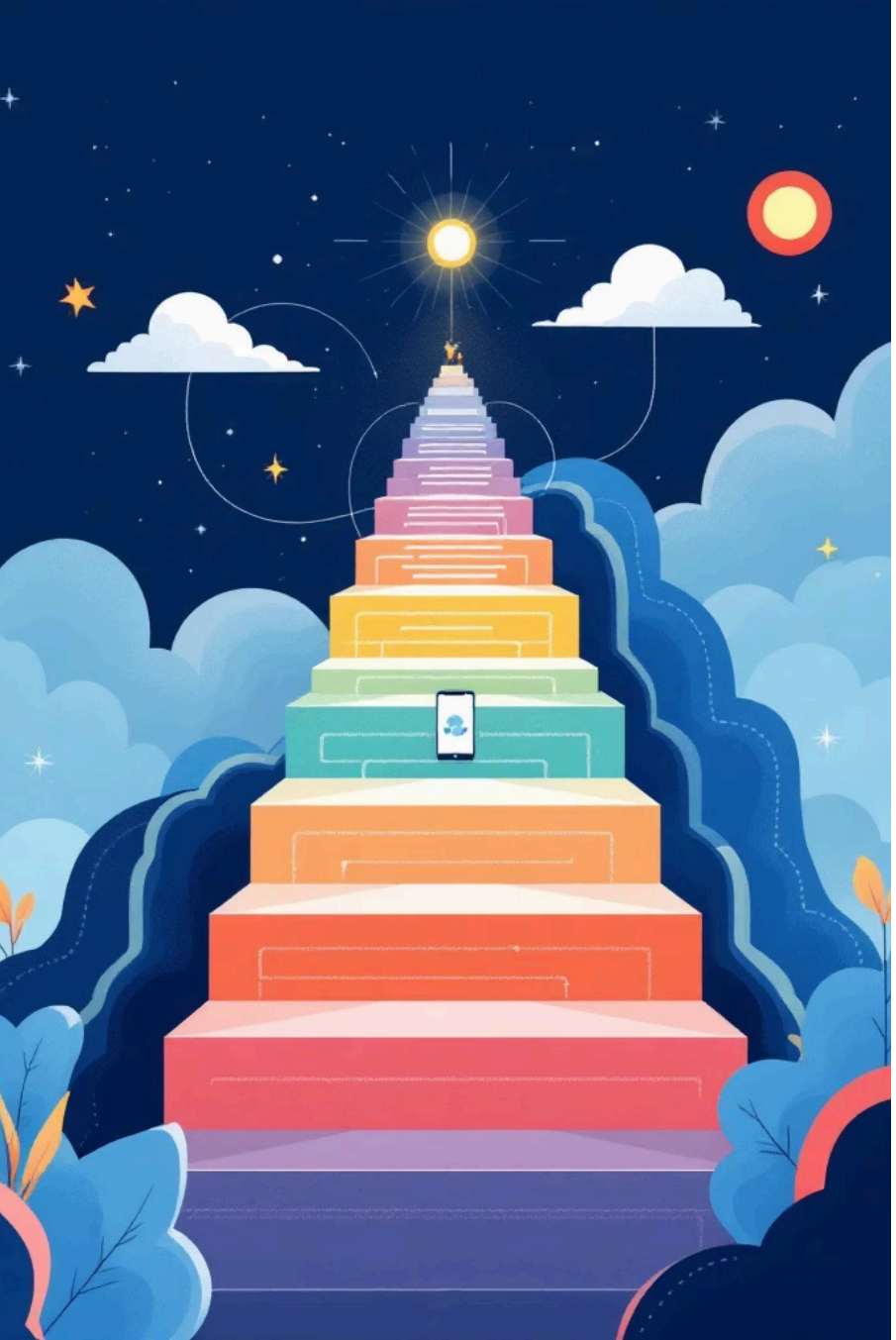
GSS



Transform Your Career Journey

This bootcamp bridges the gap between Python fundamentals and cutting-edge Generative AI development

1

Python Fundamentals

Master core programming concepts and best practices

2

Hands-On Projects

Build real AI applications with industry tools

3

AI Expertise

Deploy end-to-end Generative AI systems

Two-Phase Learning Journey

Phase 1 – Foundations

Python fundamentals

Text processing

Traditional NLP techniques

1

2

Phase 2 – Advanced GenAI

Open-source LLMs

Prompt engineering

LangChain framework

RAG systems

AI Agents

Streamlit web apps

Each phase builds upon the previous, ensuring solid understanding before advancing to complex AI concepts

Core Skills You'll Master

Python Programming

Fundamentals and best practices

API Integration

Working with RESTful services

Hugging Face Models

Deploying pre-trained models

Prompt Engineering

Optimizing AI interactions

Git & GitHub

Version control essentials

AI Applications

Building intelligent systems

Real Career Outcomes

Python Applications

Build robust, scalable applications using industry-standard practices

Create APIs

Develop RESTful APIs for seamless system integration

End-to-End GenAI Systems

Deploy complete Generative AI solutions from concept to production

Chatbots & Agents

Design intelligent conversational AI assistants

AI Web Apps

Deploy mini AI-powered web applications with Streamlit

Portfolio Building

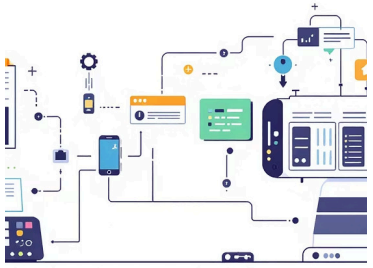
Add impressive AI projects to resume and GitHub profile

Exciting Projects Await

Build impressive applications that showcase your new AI skills

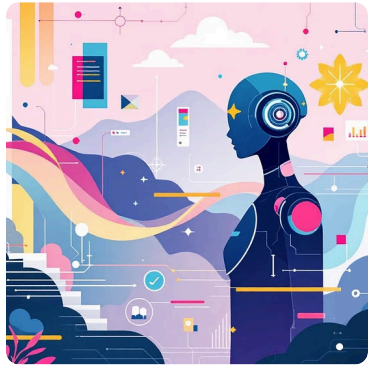
AI Chatbot

Build intelligent chatbots using Groq, Gemini, or DeepSeek APIs



Document Chatbot

Create RAG-based chatbots that answer questions from your documents



Research Agent

Develop autonomous AI agents that conduct research and gather information

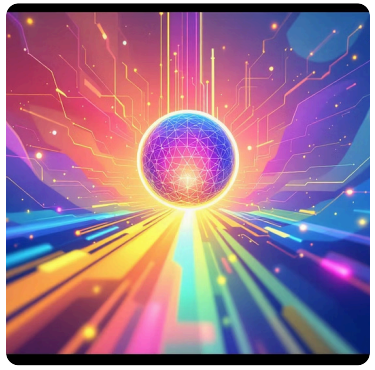
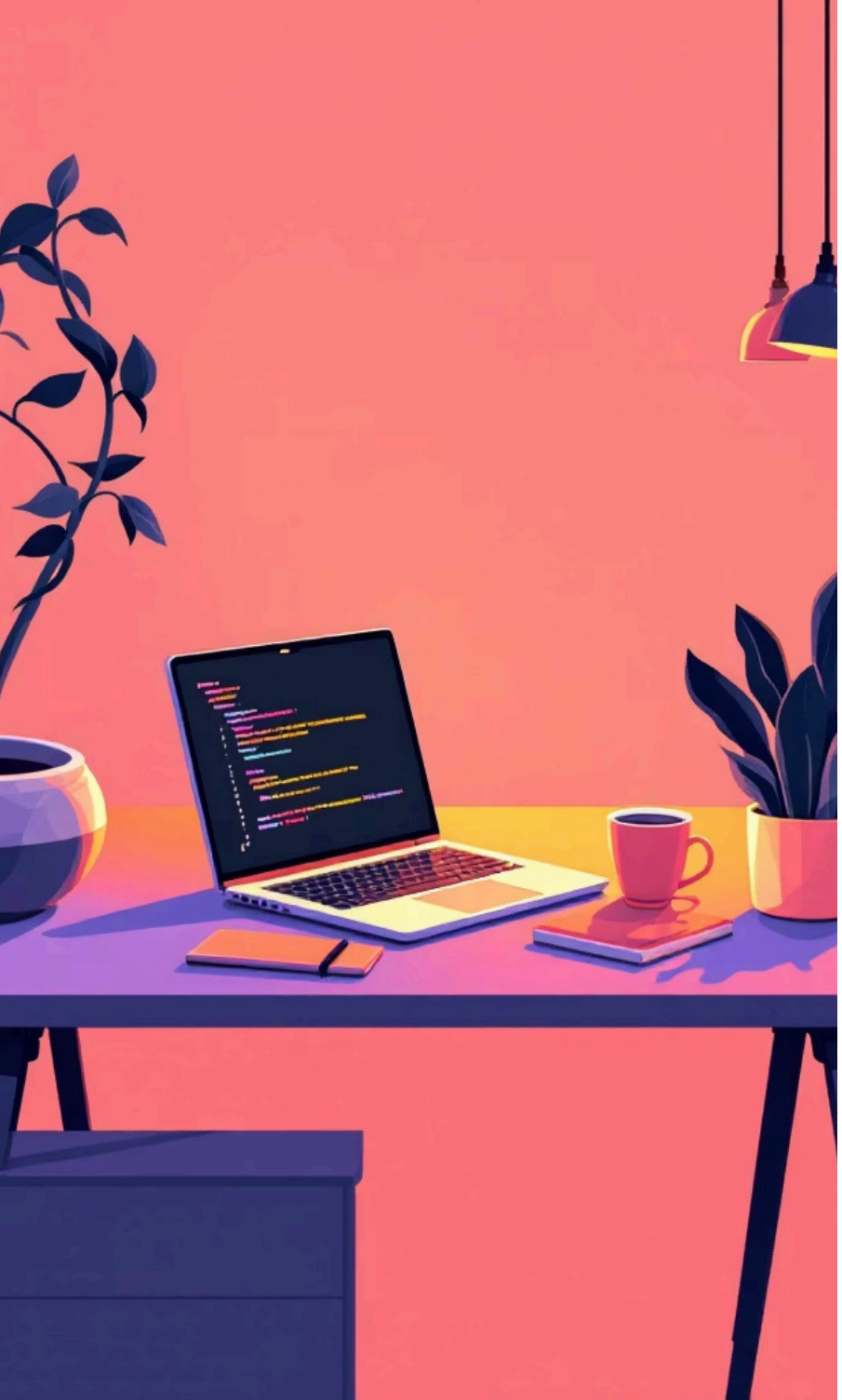


Image Generation

Generate stunning images using Hugging Face's powerful models



Today's Learning Plan

Python Basics & Setup

Python Installation

Set up your development environment

Google Colab & Jupyter

Master notebook-based development

Core Concepts

Variables, data types, and strings

String Manipulation

Essential text processing techniques

Today we build the foundation for AI. These fundamental skills are essential for everything that follows.

Google Colab & Jupyter Notebooks

What is Google Colab?

A free, cloud-based Jupyter notebook environment powered by Google

- No installation required
- Access to GPUs and TPUs
- Perfect for AI development
- Shareable and collaborative

Why Use Notebooks?

Notebooks combine code, output, and documentation in one place, making them ideal for experimentation and learning

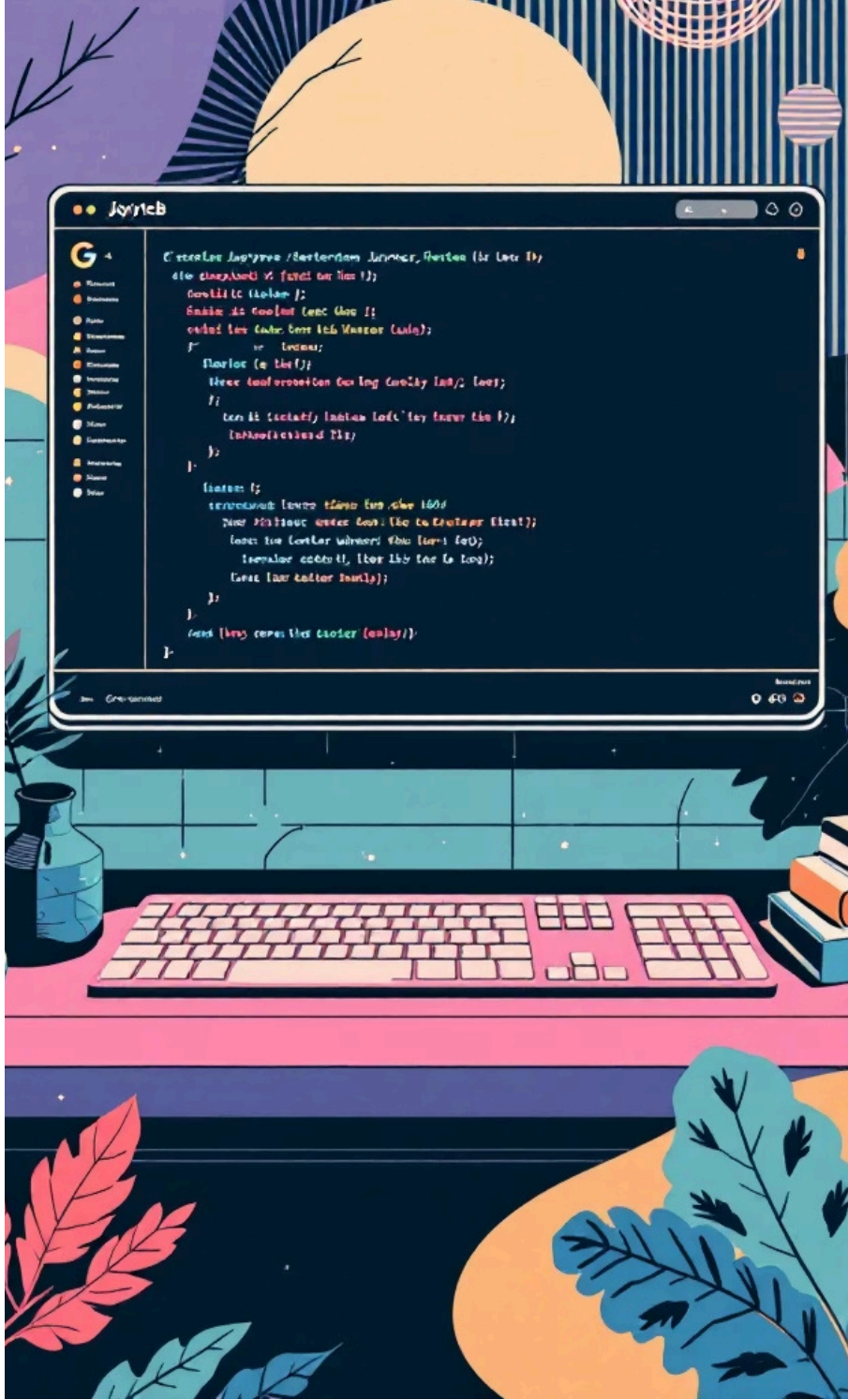
.py File vs Notebook

.py Files: Pure Python code, run entirely at once, traditional programming approach

Notebooks: Cells run independently, immediate feedback, perfect for exploration and iterative development

Running Code Cells

Execute code in chunks, see results instantly, iterate quickly, perfect for learning and experimentation



What is Python?

High-Level Language

Easy to read and write with clear, concise syntax

AI & ML Powerhouse

Widely used in artificial intelligence and machine learning

Industry Standard

Used by major tech companies including Google, Meta, and Netflix

Python's simplicity makes it perfect for beginners, while its power and flexibility make it a favorite among professional developers building AI systems